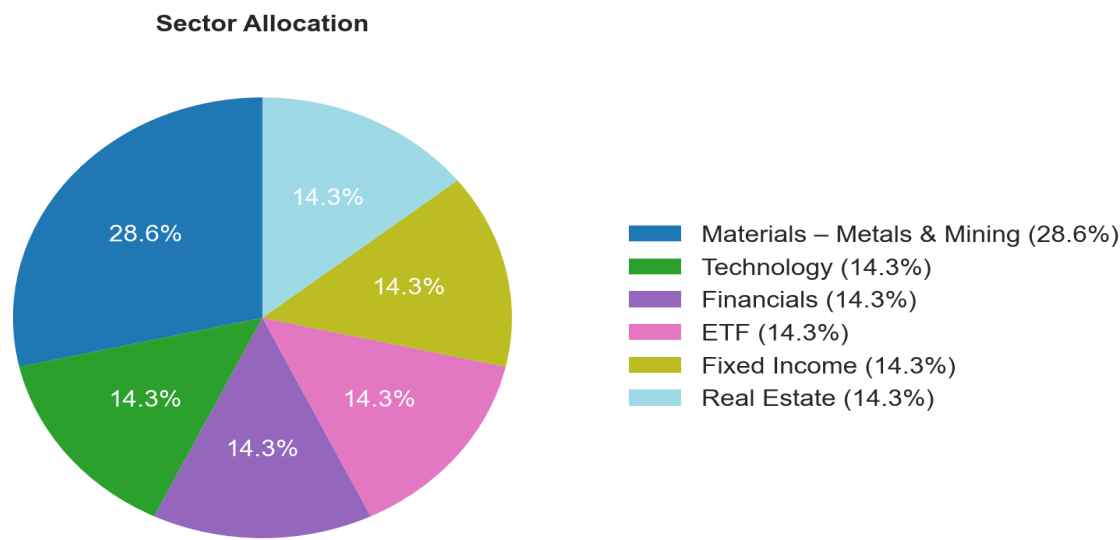


Portfolio rotation report

Date of extraction	2025-12-06 13:53
Data window	2020-01-01 → 2024-10-01
Frequency	Monthly — 58 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	AGG, DBC, GLD, SPY, VNQ, XLF, XLK
Indicators	Short term: 5 months volatility, 9 months return, 12 months momentum Long term: 17 months volatility, 22 months return, 60 months momentum

1. Portfolio Composition

- XLK (Technology Select Sector SPDR Fund — Technology)
- XLF (Financial Select Sector SPDR Fund — Financials)
- GLD (SPDR Gold Shares — Materials – Metals & Mining)
- DBC (Invesco DB Commodity Index Tracking Fund — Materials – Metals & Mining)
- SPY (SPDR S&P; 500 ETF — ETF)
- AGG (iShares Core US Aggregate Bond ETF — Fixed Income)
- VNQ (Vanguard Real Estate ETF — Real Estate)



Macro Profile Insight

Macro coverage

- Balanced diversification: exposure to most macroeconomic drivers.

Sectorial biases

- Concentration risk: strong bias toward large caps, stability achieved at the cost of diversity.

Structural properties

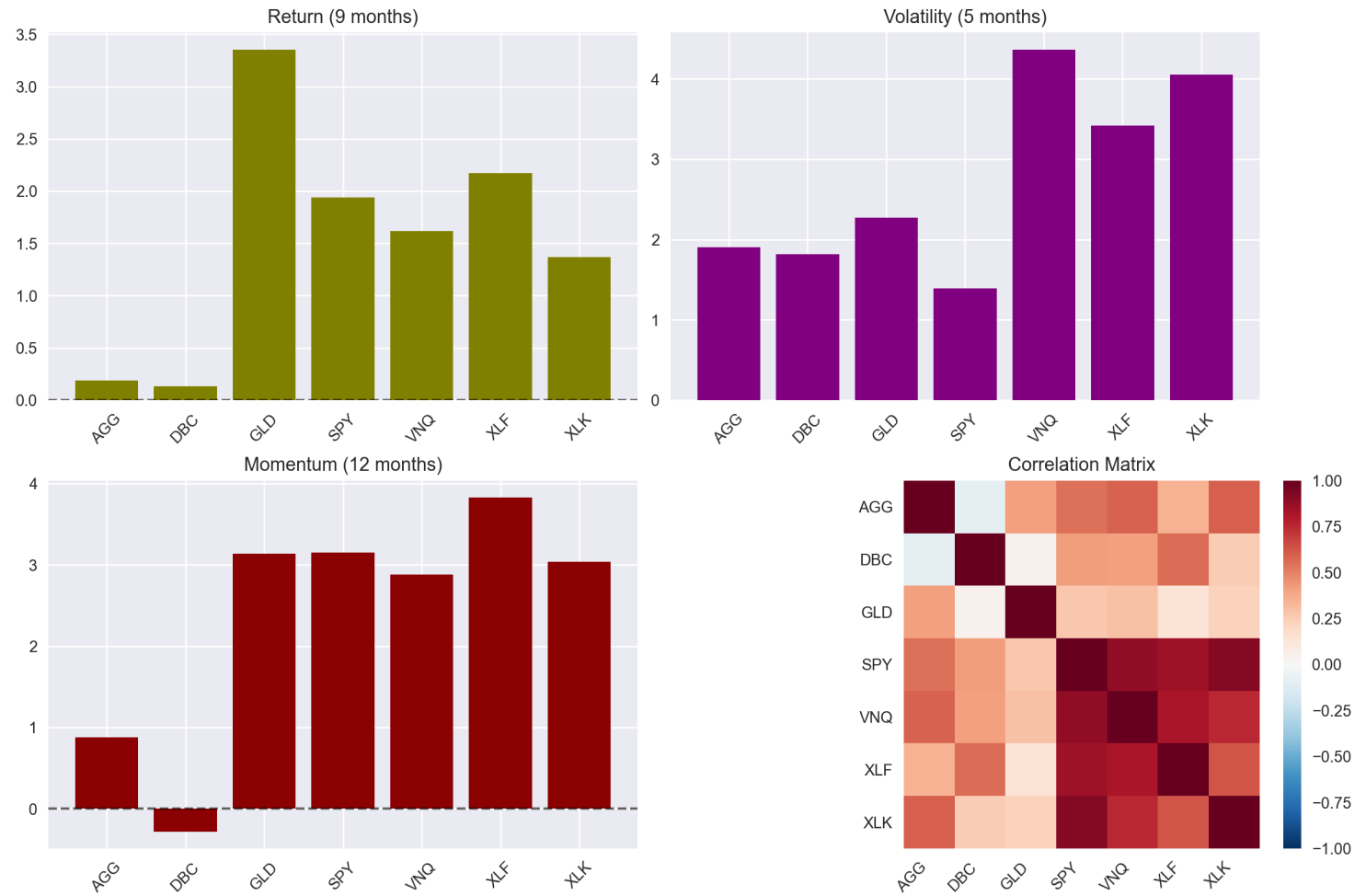
- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names.
- Defensive beta tilt: lower systematic risk profile.
- Balanced volatility structure: risk evenly distributed across holdings.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization

Sector Performance Dashboard



Top Positive Correlations

81.0% of portfolio

Pair	Correlation
SPY / XLK	0.92
SPY / VNQ	0.89
SPY / XLF	0.85
VNQ / XLF	0.82
VNQ / XLK	0.75

Top Neutral Correlations

19.0% of portfolio

Pair	Correlation
DBC / GLD	0.04
AGG / DBC	-0.09
GLD / XLF	0.14
GLD / XLK	0.23

Overall, average cross-sector correlation was 0.47, ranging from -0.09 to 0.92. This suggests elevated co-movement — diversification benefits are partially reduced. Correlation dispersion is wide, showing sharp contrasts between sectors. Strongest co-movement between SPY and XLK (0.92); weakest between AGG and DBC (-0.09). Mixed regime: both market-wide and sector-specific forces are at play.

3. Key Performance Indicators

3.a Current Regime Snapshot

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
AGG	0.19	1.90	0.103	0.88	neutral	subdued	thin	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, under limited liquidity.
DBC	0.13	1.82	0.071	-0.28	weak reversal	normal	illiquid	mixed short-term dynamics, early reversal signs, short-term momentum starting to diverge, normal volatility regime, under limited liquidity.
GLD	3.35	2.27	1.476	3.14	strongly coherent	elevated	thin	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, rising volatility — tension building, under limited liquidity.
SPY	1.94	1.39	1.396	3.15	strongly coherent	subdued	liquid	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, low volatility — calm market tone, with ample liquidity.
VNQ	1.62	4.36	0.372	2.88	mildly coherent	subdued	illiquid	positive performance with solid momentum, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.
XLF	2.17	3.42	0.634	3.83	strongly coherent	subdued	thin	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, low volatility — calm market tone, under limited liquidity.
XLK	1.37	4.05	0.338	3.04	strongly coherent	normal	thin	positive performance with solid momentum, signal remains strongly coherent with long-term momentum, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
GLD	3.35
XLF	2.17
SPY	1.94
VNQ	1.62
XLK	1.37

Top by Volatility

Ticker	Volatility(%)
VNQ	4.36
XLK	4.05
XLF	3.42
GLD	2.27
AGG	1.9

Top by Momentum

Ticker	Momentum(%)
XLF	3.83
SPY	3.15
GLD	3.14
XLK	3.04
VNQ	2.88

Last by Average Return

Ticker	AvgReturn(%)
DBC	0.13
AGG	0.19
XLK	1.37
VNQ	1.62
SPY	1.94

Last by Volatility

Ticker	Volatility(%)
SPY	1.39
DBC	1.82
AGG	1.9
GLD	2.27
XLF	3.42

Last by Momentum

Ticker	Momentum(%)
DBC	-0.28
AGG	0.88
VNQ	2.88
XLK	3.04
GLD	3.14

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
AGG	-0.13	-8.80	-0.09	0.91	0.03	8.66 M	stable performance, momentum fading, volatility normalizing, within defensive profile, under thin liquidity regime.
DBC	0.35	-0.29	-0.217	0.48	0.28	453.3 K	stable performance, momentum fading, volatility collapsing, within defensive profile, under thin liquidity regime.
GLD	1.59	2.42	0.919	0.66	0.04	9.35 M	acceleration in returns, momentum surging, volatility compressing, within defensive profile, under thin liquidity regime.
SPY	0.06	1.84	1.108	0.39	1.00	72.91 M	stable performance, momentum surging, volatility collapsing, aligned with market dynamics, under robust trading volume.
VNQ	0.78	8.36	0.308	0.78	0.59	3.60 M	stable performance, momentum surging, volatility compressing, within defensive profile, under thin liquidity regime.
XLF	0.73	2.92	0.978	0.84	0.84	36.84 M	stable performance, momentum surging, volatility compressing, aligned with market dynamics, within normal liquidity conditions.
XLK	-1.22	1.02	1.503	0.83	1.33	16.33 M	loss of momentum in returns, momentum stable, volatility compressing, amplifying market trends, within normal liquidity conditions.

■ High-impact observations

For **XLK**: *Technical rebound* — improving momentum while returns are still limited. (avg ann. return 1.4%, momentum 3.0%, vol 4.0%)

■ Moderate-impact observations

For **SPY, XLF**: *Technical rebound* — improving momentum while returns are still limited. (avg ann. return 2.1%, momentum 3.5%, vol 2.4%)

■ Light-impact observations

For **AGG, VNQ**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.9%, momentum 1.9%, vol 3.1%)

For **DBC, GLD**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return 1.7%, momentum 1.4%, vol 2.0%)